NP-Completeness Homework

1 Complexity Classes

Problem 1.1. State the definitions of the following sets: P, NP, **NP-Complete**, and **NP-Hard**

P = set of problems SOLVABLE in polynomial time.
NP = set of problems VERIFIABLE in polynomial time.
NP-Hard $= \{Q : \forall q \in NP, q \leq_p Q\}$
NP-Complete = NP $\cap$ NP-Hard

Problem 1.2. Suppose that P_1 and P_2 are problems and

- Problem P_1 reduces to P_2 in polynomial time.
- Problem P_2 can be **solved** in polynomial time.

What can we conclude about P_1? Why?
Your answer may be "we can conclude nothing".

$P_1 \leq_p P_2$; $P_2 \in P$
$\therefore P_1 \in P$

Intuition: "$P_1 \leq$ polynomial" means $P_1 \in O(\text{polynomial})$

Problem 1.3. Suppose that P_1 and P_2 are problems and

- Problem P_2 reduces to P_1 in polynomial time.
- Problem P_2 can be **solved** in polynomial time.

What can we conclude about P_1? Why?
Your answer may be "we can conclude nothing".

We can conclude nothing.
$P_1 \leq_p P_2$
$P_1 \in P$

Problem 1.4. Prove that the following problems are in **NP** by

(a) Writing an algorithm which **verifies** a solution.

(b) Showing that your algorithm takes at most polynomial time.

Your solution should be a function which takes a "problem" and a "proposed solution" as input and returns true if the proposed solution solves the problem.

Island Clique Given a weighted graph $G = (V, E)$ and integer K is there a clique of size at least K in G such that the clique is totally disconnected from the rest of the graph?

Given Graph G
Integer K
Set C

$\Theta(1)$ 1. Check $|C| \geq K$

$O(n^3)$ 2. Check C is a clique

```
for u ∈ C
    for v ∈ C
        check if (u,v) ∈ G.E
```

$O(n^3)$ 3. Check C is disconnected.

```
for u ∈ C
    for v ∈ (G.V - C)
        check if (u,v) ∈ G.E
```

Thus, Island Clique $\in$ NP

Word chain Given a list L containing n strings and an integer K, does L contain a subset C of size k such that for $i = 1, 2, \ldots, k-1$,

$$\text{Last symbol of } C[i] = \text{First symbol of } C[i+1]$$

For example, if $L = [\text{"cat"}, \text{"taco"}, \text{"plant"}, \text{"dog"}, \text{"orthogonal"}]$ and $k = 3$, then $C = [\text{"cat"}, \text{"taco"}, \text{"orthogonal"}]$ is a valid subset for the problem, but if $k = 4$ then there is no valid subset.

Note: I think this is an **NP**-Complete problem, but have thus far been unable to prove it.

Given list L, int K, set C

1. Verify $|C| = K$. $\Theta(1)$
2. Verify $C \subseteq L$ for $i=1$ to K check if $C[i] \in L$ $O(Kn)$
3. for $i = 1$ to $K-1$
   check if $LAST(C[i]) = FIRST(C[i+1])$ $O(K)$

Thus, we can verify in $O(nK)$ time.

Therefore in NP.

Problem 1.5. Prove that the following problems are in **P** by

(a) Writing an algorithm which **solves** the problem.

(b) Showing that your algorithm takes at most polynomial time.

Closest points Given a collection of points $S = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\} \subseteq \mathbb{R}^2$ in the xy-plane and a distance $\delta > 0$, is there a pair of points with distance $\leq \delta$?

If you want an *extremely* difficult coding challenge, try to solve this recursively using divide and conquer in $\Theta(n \log(n))$ time.

Given S, δ

For each pair of points p_i, p_j

check if $\text{dist}(p_i, p_j) < \delta$

Solved in $O(n^2)$ time $\Rightarrow$ P.

ALTERNATIVE

```
function ClosestPoints(S, δ)
    for i=1 to n
        for j=i+1 to n
            d = distance((x_i,y_i),(x_j,y_j))
            if d ≤ δ
                return true
            end if
        endfor
    endfor
```

$O(n^2)$

2-Coloring Given a graph G, is there a coloring of the vertices with 2 colors so that no two adjacent vertices have the same color?

Hint: You do not need to write code, you can describe a procedure that takes at most polynomial time without the specificity of code.

Given Graph G,

while $\exists$ vertex with no color ← $O(n)$ iterations

1. Perform BFS on some uncolored vertex v ← $O(n+m)$
2. Color all vertices that are an even distance BLUE
3. Color all vertices that are an odd distance RED

Solved in $O(n^2+nm)$ time $\Rightarrow$ P.

ALTERNATIVE

```
function TwoColor(G)
    U = G.V
    while U ≠ ∅
        let v = some vertex in U
        Perform BFS(G,v)
        foreach vertex x ∈ G.V
            if x.d is even
                x.color = blue
            else if x.d is odd
                x.color = red
            endif
            if x.d ≠ ∞
                remove x from U
            endif
            x.d = ∞
        endfor
    endwhile
```

```
for each edge (u,v) ∈ G.E
        if u.color == v.color
            return false
        endif
endfor
return true.
```

2 Common NP-C Problems

Problem 2.1. Give a truth assignment which satisfies the following Boolean expression,

$$\varphi = (x_2 \vee \overline{x}_3 \vee x_4) \wedge (\overline{x}_1 \vee x_2 \vee x_4) \wedge (\overline{x}_2 \vee \overline{x}_3 \vee \overline{x}_4) \wedge (\overline{x}_1 \vee \overline{x}_3 \vee x_4) \wedge (\overline{x}_1 \vee x_3 \vee \overline{x}_4)$$

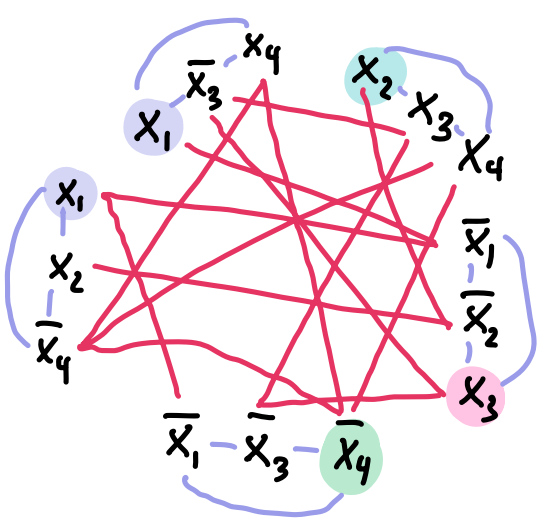
$x_2 = T$
$x_3 = F$
$x_4 = F$
$x_1 = T$ or F

Problem 2.2. Consider the following Boolean expression

$$\varphi = (x_1 \vee x_2 \vee \overline{x}_4) \wedge (x_1 \vee \overline{x}_3 \vee x_4) \wedge (x_2 \vee x_3 \vee x_4) \wedge (\overline{x}_1 \vee \overline{x}_2 \vee x_3) \wedge (\overline{x}_1 \vee \overline{x}_3 \vee \overline{x}_4)$$

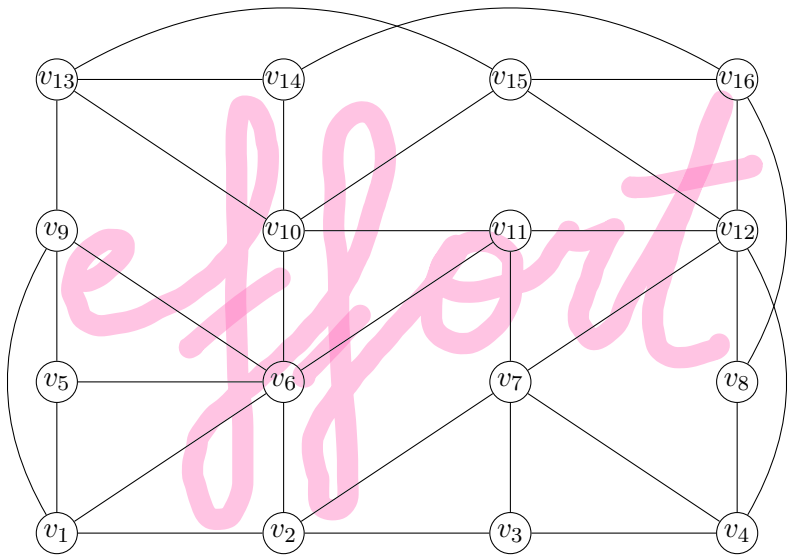
(a) Give the graph G corresponding to φ in the reduction of 3-SAT to the Independent Set problem as we performed in the notes.

(b) Give an independent set of size 5 in G and a corresponding truth assignment which satisfies φ.



$x_1 = T$
$x_2 = T$
$x_3 = T$
$x_4 = F$

Problem 2.3. For the given graph,



(a) What is the largest independent set that you can find?

(b) What is the smallest vertex cover that you can find?

(c) What is the largest clique you can find?

(d) Can you find a Hamiltonian Path from v_1 to v_{16}?

Problem 2.4. Consider the following two problems,

Solvable Sudoku Given a partially filled 9×9 Sudoku board, is there a solution to the puzzle? Recall that a Sudoku is solved if

(a) Each row must contain the digits 1-9 without repetition.

(b) Each column must contain the digits 1-9 without repetition.

(c) Each of the 9 sub-boxes of size 3×3 of the grid must contain the digits 1-9 without repetition.

9-Colorable Graph Given a graph G with some colors assigned, can G be properly colored with 9 colors?

Explain how you could reduce the Solvable Sudoku problem to the 9-Colorable Graph Problem. That is, given a partially filled Sudoku puzzle, construct a graph G such that the Sudoku is solvable if and only if the graph can be colored with 9 colors.

If you want a challenge, generalize this to an $n^2 \times n^2$ Sudoku and an n^2 Colorable graph.

If you want even more of a challenge, make this a formal proof!

Idea: Create a vertex for each cell.

Assign a color to each digit 1-9.

Draw edges between the following vertices:

All vertices in the same cell row.

All vertices in the same cell column.

All vertices in the same cell block.

For each digit present in the original puzzle, color the corresponding vertex the appropriate color.

[Y ⇒ Y] Suppose the sudoku has a solution. Assign colors to the vertices based on the solution. We must now argue that this coloring is valid.

1. Since each row contains only 1 instance of 1-9, it does not have any color violating edges.
2. Same for each column.
3. Same for each block.

This accounts for every edge in the graph.

Thus, G is properly colored.

[Y ⇐ Y] This argument is IDENTICAL, just reversed.

Coloring ⇒ assign 1-9 ⇒ each row/col./block has 9 colors ⇒ valid sol'n.

To make valid for $n^2 \times n^2$ ⇒ replace all 9's with n^2's.

3 Reductions

Problem 3.1. Prove that the following problem is in **P** by reducing it to Minimum Spanning Tree.

Maximum Spanning Tree Given a weighted graph $G = (V, E)$ and a cost C, is there a spanning tree T of G such that the total cost of T is $\geq C$?

$O(n+m)$ Create a graph $G' = (V, E)$ where all the edge weights multiplied by -1, set $C' = -C$.

$O((n+m)\log(n))$ Run Prim's on $\bar{G}$.

$O(n)$ Check if weight of tree $\leq C'$. $\leftarrow \sum_e -w(e) \leq -C$

$\downarrow$

$\sum_e w(e) \geq C$

Thus, Max ST $\in$ P

Problem 3.2. Prove that the following problem is in **NP**.

Hint: This can be *extremely* easy.

Maximum Spanning Tree Given a weighted graph $G = (V, E)$ and a cost C, is there a spanning tree T of G such that the total cost of T is $\geq C$?

Max Spanning Tree $\in$ P $\Rightarrow$ Max Spanning Tree $\in$ NP

Problem 3.3. Consider the following two problems,

Subset Sum Given a set $X = \{x_1, x_2, \ldots, x_n\}$ and integer K, is there a subset $Y \subseteq X$ such that $\sum_{x \in Y} x = T$.

Optimal Change Given a set of coins $C = \{c_1, c_2, \ldots, c_n\}$ and integers S, K, is there a multiset (a set that can contain duplicates) $M \subseteq C$, such that $\sum_{x \in M} x = K$ and $|M| \leq S$?

Explain why these problems are similar.

Both problems want to find a subset of values that sum to a target.

Optimal change is just subset sum with duplicates.

NOTE: I am WAY more formal than you need to be.

Problem 3.4. Consider the following two problems,

3-coloring Given a graph G, can G be properly colored with 3 colors?

3-clique Cover Given a graph G, can G be partitioned into 3 disjoint cliques such that the cliques cover the graph? That is, does there exist 3 sets of vertices A, B, C such that $A \cap B = A \cap C = B \cap C = \varnothing$ and $A \cup B \cup C = G.V$.

(a) Prove that the 3-clique Cover problem is in NP.

(b) Prove that the 3-coloring Problem reduces to the 3-clique Cover Problem in polynomial time.

(c) What can we now conclude about the 3-clique Cover problem?

a) Given a graph G and 3 sets A, B, C

Verify $A, B, C \subseteq V$ — foreach $v \in A$ check $v \in V$ } repeat for B, C — $O(n^2)$

Verify $A \cap B = A \cap C = B \cap C = \varnothing$ — foreach $a \in A$ check if $a \in B$ check if $a \in C$ — $O(n^2)$

Verify $A \cup B \cup C = V$ — foreach $v \in V$ check if $v \in A$ check if $v \in B$ check if $v \in C$ — $O(n^2)$

Verify each set is a clique — repeat for B, C { foreach $a_1 \in A$ foreach $a_2 \in A$ check $(a_1, a_2) \in E$. — $O(n^2 m)$

REALLY pessimistic

---

b) Let G be an instance of the 3-coloring problem. Our idea for the reduction is if G is 3-colorable, then the colors do not touch. This is kinda the opposite of a clique. In particular, let $\bar{G}$ be the graph complement of G and an instance of the 3-clique cover problem.

[Y ⇒ Y] Suppose G is 3-colorable (assume the colors are c_1, c_2, c_3).

Let $A = \{v \in G.V : v.\text{color} = c_1\}$.

Let $B = \{v \in G.V : v.\text{color} = c_2\}$.

Let $C = \{v \in G.V : v.\text{color} = c_3\}$.

Since G is 3-colorable, $\forall x, y \in A$ if $x \neq y$ there is no edge $(x, y) \in G.E$.

Since $\bar{G}$ is the graph complement, $(x, y) \in \bar{G}.E$ by definition.

That is, $\forall x, y \in A$, if $x \neq y$ then $(x, y) \in \bar{G}.E$.

Thus, A is a clique of $\bar{G}$. Similar for B and C.

Since A, B, C come from the coloring, $A \cup B \cup C = V$ and are disjoint.

Therefor A, B, C form a 3-clique cover of $\bar{G}$.

[Y ⇐ Y] Suppose $\bar{G}$ has a 3-clique cover A, B, C.

$\forall a \in A$, set $a.\text{color} = \text{RED}$

$\forall b \in B$, set $b.\text{color} = \text{BLUE}$

$\forall c \in C$, set $c.\text{color} = \text{GREEN}$

Since A is a clique $\forall x, y \in A$, if $x \neq y$ then $(x,y) \in \bar{G}.E$.

Thus, by definition of graph complement, $(x,y) \notin G.E$.

That is, if $x, y \in V$ are both RED, then $(x,y) \notin G.E$.

Similar for BLUE and GREEN.

Furthermore, since $A \cup B \cup C = V$, all nodes have at least one color. Since A, B, C are disjoint we also know each node has <u>ONLY</u> one color.

There G is 3-colorable.

The reduction takes $O(n)$ time to copy the vertices and $O(n^2)$ time to generate the edges of the complement.

Thus,

$$\boxed{\text{3-coloring} \leq_p \text{3-clique cover}}$$

c) Therefore, 3-clique cover is NP-C.